A certificate-based audit of the fundamental-group and framing computations in Wuebben's proposed exotic $S^2 \times S^2$ construction

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Abstract

We audit the fundamental-group and framing calculations in Wuebben's proposed exotic $S^2 \times S^2$ construction. Four explicit, hash-identified finite presentations are proved trivial by replayable derivation certificates under a published certificate specification. From the marked data we define an explicit audit manifold V_* and prove, in separate geometric steps, that those presentations are presentations of $\pi_1(V_*)$; this makes the common whiskers, meridians, product longitudes, drilled transport relation, and product-to-Lagrangian framing bridge inspectable. We then isolate four source-identification assumptions S1–S4 under which V_* is Wuebben's intended fixed surgery member. Thus V_* is simply connected, while the transfer of that conclusion to Wuebben's member is explicitly conditional on S1–S4. Conditional also on twenty-five named external results, the three manifold conclusions of Wuebben's paper follow. A reported contrary fundamental-group computation is stated separately and bears on the source comparison, not on the finite certificate theorem. The theorems are Wuebben's; this note claims only the audit.

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Status of the claims. The four hash-identified presentations are trivial, and the audit-defined manifold V_* is simply connected, relative only to the displayed geometric proofs, named standard topology inputs, and checker correctness. The further assertion that V_* is Wuebben's intended fixed member uses the four source-identification assumptions S1–S4 stated in Section 3.1. The three manifold conclusions also use the cited classification, symplectic, and Floer-theoretic results. An unpublished contrary fundamental-group calculation reported by Wuebben remains unreconciled and bears precisely on the conditional source comparison.

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1 Introduction

For the construction of [1], the theorem-critical step is the fundamental-group computation. The manifolds there rest on a specified Lidman–Piccirillo piece V, obtained by two Luttinger surgeries [8, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [2]. Write σ for the free orientation-reversing involution of $\partial V = S^3_0(Q)$, Q the square knot, $Z = V \cup_\sigma V$ for the symplectic double, $W = V/\sigma$ for the quotient, and $Z'' = V \cup_{f\circ\sigma} V$ for the Lidman–Piccirillo regluing. The paper's three theorems are: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to Kawauchi's manifold B, and (B, W) is a homeomorphic, non-diffeomorphic pair distinguished by the sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. All three rest on the simple connectivity of V. Wuebben fixes one geometric manifold: his $(m, n) = (0, 0)$ member, using two geometric $+1$ Luttinger surgeries and the y_1-based Ax half-drift on T_α. The exponents ϵ_A, ϵ_B in his relation sheet are orientation and presentation conventions, not four geometric parametrizations; he allows either sign for each. We certify all four convention specializations, so in particular the relator sheet for the fixed V is included. The other three specializations are convention-robustness checks, not claims about three further manifolds. Wuebben's own 4,096-system grid varies, besides these two exponents, the three meridian signs of his transport corrections, the left/right placement of those corrections, the choice of conjugating arc for the Bs correction, and the arc, sign, and placement of the T_β push-off. In the rebuilt model none of these is a free choice: the corrections, basings, and push-off are computed as literal certified loops (Section 4), so only ϵ_A, ϵ_B remain as conventions, and the four certificates cover his grid at the fixed member.

This note reports an independent implementation of that step, a machine audit of the framing lemma that connects the group theory to the symplectic geometry, and an explicit proof chain from simple connectivity to the three theorems. Nothing in the implementation reuses the paper's coordinates, code, or intermediate data: the geometry was rebuilt from the paper's stated marked-fiber data alone, and the paper's displayed relations, sign tables, and Table 1 filling words were then *compared against* the rebuild rather than assumed. Every mechanically comparable item agreed, and the independent raw-paper extractor found no dictionary discrepancy. That agreement does not prove what the author's prose and figures are intended to denote. Section 3.1 therefore states the remaining semantic assertions as S1–S4 instead of hiding them inside a geometric theorem. The downstream deductions are Wuebben's, and this note reproves none of the classification, symplectic, or Floer-theoretic theorems they invoke; what it adds is that each deduction is written out, each theorem invoked is stated with its hypotheses, and each hypothesis is discharged by a named certificate, computation, source assumption, or earlier step.

For the notation below, "$n = 0$" means Wuebben's Ax half-drift label on T_α; the adjacent "$n = 1$" control replaces Ax by Ar^{-1}. The letters M, N denote based meridians of T_α, T_β, while A, B are the two base generators.

The argument has four deliberately separate layers: a finite theorem about explicit presentations; a geometric identification with the explicitly defined V_*; a source comparison, conditional on S1–S4, with Wuebben's intended member; and consequences conditional on named external results. The first is formalized next. The audit-model geometry is the center of Sections 3–4; no source assumption enters its simple-connectivity conclusion.

2 The finite certificate theorem

Theorem 2.1 (finite-certificate statement)**.** Let `S` be the byte string

`verification/luttinger/sealed_transport/r_presentations.json`,

whose SHA-256 digest is

`1e5fc8cb6872a25fa45a227c1e1c207b288c861a3998fe3aae8722ea510c852c`.

Read a positive integer j in `S` as the generator g_j and a negative integer $-j$ as g_j^{-1}. The top-level fields `ngens=3` and `relators` define a 3-generator, 78-relator presentation Q. Take the four entries of the top-level array `paper_fillings` whose `half_drift` field is `"n0_y1"`; these are entries 0–3, and their (`sign_a`, `sign_b`) fields run over $(+1,+1)$, $(+1,-1)$, $(-1,+1)$, $(-1,-1)$ in that order. Append the two word lists of such an entry, of lengths 94 and 132, to the 78 relators of Q, and call the resulting 3-generator, 80-relator presentation $P_{\epsilon_A,\epsilon_B}$ for $\epsilon_A =$ `sign_a` and $\epsilon_B =$ `sign_b`. Every one of these four explicitly defined presentations presents the trivial group.

Definition 2.2 (letters, words, and equation keys)**.** Fix generators $g_1, \dots, g_n$. Source relators are signed words in $\{\pm 1, \dots, \pm n\}$, with j denoting g_j. Certificate records use the monoid alphabet $L_n = \{1, \dots, 2n\}$ and the decoding

$$\sigma(2j-1) = j, \qquad \sigma(2j) = -j.$$

Thus the required inverse-letter table is $(2, 1, 4, 3, \dots, 2n, 2n-1)$. If w is a signed word, write $\bar{w}$ for its reverse with all signs changed, $\mathrm{fr}(w)$ for free reduction, and $\mathrm{cr}(w)$ for the word obtained by additionally removing inverse first/last pairs until none remain. Define

$$\mathrm{ck}(w) = \min_{\mathrm{lex}}\{\text{cyclic rotations of } \mathrm{cr}(w) \text{ and } \overline{\mathrm{cr}(w)}\},$$

with $\mathrm{ck}(1) = 1$. For certificate words u, v put

$$E(u,v) = \mathrm{ck}\big(\sigma(u)\overline{\sigma(v)}\big).$$

Equality of equation keys therefore means equality after free and cyclic reduction, cyclic conjugacy, and inversion. Each operation preserves the normal closure of the equation word.

Definition 2.3 (literal rewrite trace)**.** Suppose records $R_0, \dots, R_{i-1}$ have been accepted, with $R_r = (\ell_r, \rho_r)$. A trace on a word w is a list of pairs (r, p) with $r < i$ and $0 \le p \le |w|$. At that step the contiguous subword beginning at p must equal ℓ_r literally, and it is replaced by ρ_r. No cyclic matching, inversion, or implicit free reduction occurs inside a trace. Write $T(w)$ for its final word.

Definition 2.4 (certificate grammar)**.** A `luttinger-kbmag-proof-v1` certificate is a finite ordered list of records (ℓ_i, ρ_i), together with source-binding metadata and identity roots. Every side is a word over L_n. The case index must be an integer in the range of the source filling array. The source digest, case index and slug, generator count, full relator list, and inverse table must agree exactly with the hash-bound input. A record has exactly one of four proof forms.

(i) An *inverse axiom* has $\rho_i = 1$ and $\ell_i = a\,\iota(a)$ for the required inverse table ι.

(ii) An *input-relator record* names an in-range source relator q_k and satisfies $E(\ell_i, \rho_i) = \mathrm{ck}(q_k)$.

(iii) An *overlap record* names parents $a, b < i$, an integer offset d with $-|\ell_b| < d < |\ell_a|$, and two traces. The two parent left sides must agree literally on the resulting nonempty overlap. Let w be the shortest word containing both occurrences. Replacing the occurrence of ℓ_a by ρ_a gives one branch w_a, and replacing ℓ_b by ρ_b gives the other w_b. If the recorded traces give $u = T_a(w_a)$ and $v = T_b(w_b)$, validity requires $E(\ell_i, \rho_i) = E(u, v)$.

(iv) A *change record* names an old record $o < i$ and traces on its two sides. Their literal outputs must equal the separately stored `reduced_left` and `reduced_right` words, say u, v, and validity requires $E(\ell_i, \rho_i) = E(u, v)$.

Finally the root array has length exactly $2n$; its entry for each $a \in L_n$ is an in-range record whose two sides are literally $(a, 1)$. Full-inventory mode also requires one correctly slugged certificate for each of the eight source fillings, with no repeated index or slug and with the stated presentation dimensions.

Theorem 2.5 (certificate soundness)**.** Let G be the group defined by the source relators for a selected filling. If a certificate satisfies Definitions 2.2–2.4, then G is trivial.

Proof. Induct over the records. An inverse axiom is an equality in the free group. For an input record, the equation word is a freely and cyclically reduced cyclic conjugate or inverse of a defining relator, so its two sides agree in G. Every trace replaces a subword by the other side of an earlier valid equation and therefore preserves the represented element of G.

For an overlap, both untraced branches come from one source word by applying one earlier equality, so they agree in G; their traces preserve that equality. Equality of equation keys transfers the resulting equality to (ℓ_i, ρ_i). For a change, the two sides of an earlier valid equation are separately rewritten by earlier equations, and the same equation-key argument applies. Thus every accepted record is an equality in G. The root records say literally that every generator letter and its inverse equal 1, so $G = 1$. □

Reference verification algorithm. The normative specification is Definitions 2.2–2.4. For comparison with the implementations, the complete control flow is:

```
bind source bytes, SHA-256, case, relators, generator count, inverse
table;
for each record in order:
  validate both words over L_n;
  dispatch to inverse-axiom, input-relator, overlap, or change;
  replay every trace using only earlier record numbers;
  compare the required literal outputs and equation keys;
check one literal [letter] → [] root for every letter; accept.
```

The standalone normative text is `verification/luttinger/CERTIFICATE_SPEC.md`. The Python and Ruby checkers separately implement it using only their standard libraries. They share neither source code nor runtime, but both were produced within this project and have not received an independent human line-by-line audit.

Proof of Theorem 2.1. For each of the four selected entries, the certificate bundle prepared with this version contains a certificate bound to the displayed source digest, case index, slug, complete relator list, generator count, and inverse table. The Python and Ruby implementations both accept every record and all six identity roots. The four certificates retain respectively 6672, 2470, 3803, and 3620 records in the source order $(+,+), (+,-), (-,+), (-,-)$. Hence Theorem 2.5 applies to each selected filling. Separately, two standalone checkers replay the frozen 99,860 elementary Tietze eliminations that produced `S` from its serialized raw input. No geometric interpretation of either byte string enters this proof. □

The finite theorem is now relative only to the published certificate specification and to the proposition that at least one implementation conforms to it. The rest of the paper addresses the mathematically different question of what the hash-bound presentations describe. The principal result there is Theorem 4.7, which identifies them with the explicitly defined V_*. Proposition 4.9 is the separate, S1–S4-conditional comparison with Wuebben. Theorem 5.1 records the downstream consequences only after both levels have been stated. Table 1 keeps these evidentiary levels separate.

Claim	Public artifact	What replay establishes	Remaining human input
$P_\epsilon = 1$	derivation DAG	every retained equation follows from earlier equations; every generator equals 1	checker implementation conforms to Theorem 2.5
raw to reduced presentation	frozen Tietze trace	each recorded elimination is valid and all 89 tracked words are transported	the raw presentation describes the geometric complement
peripheral pairs	simplicial paths and normal blocks	paths, common whiskers, orientations, local flatness, and normal-circle fibers	the standard PL interpretations stated in the proofs
framing agreement	exact-rational identities and finite hypotheses	constructive collar normalization, chart calculations, and first-jet independence	the Lagrangian-neighborhood theorem and the interpretation of the named push-offs inside V_*
V_* = Wuebben's fixed member	source dictionary, independent extraction, mutation controls, and author-script comparison	every mechanically comparable convention agrees	S1–S4
three manifold conclusions	dependency ledger	hashes, finite calculations, hypotheses recorded, and acyclic dependency	the cited theorems, S1–S4, and the semantic correctness of their applications

Table 1: Evidence levels. A replayed finite derivation, a checked combinatorial construction, and a dependency audit are intentionally not treated as the same kind of evidence.

3 The fixed geometric object and its independent model

We first define the object audited here without referring to a computer file or requiring the reader to reconstruct it from the target paper. Write $\Sigma_{1,1}$ for an oriented once-punctured torus with based oriented spine loops α, β.

Definition 3.1 (the audit-defined surgery manifold)**.** Let F be a closed oriented genus-two surface with an ordered five-chain (a, b, c, d, e) whose regular-neighborhood complement is the disjoint union of two disks containing marked points p, O. Let φ_0 be the orientation-preserving involution reversing the chain, fixing p, O, and acting freely on c by a half-rotation. Let

$$\psi_0 = T_a \circ T_b,$$

with T_b applied first and both twists supported away from a collar of e. Form the marked bundle R_* over $\Sigma_{1,1}$ by clutching the α- and β-mapping cylinders of φ_0 and ψ_0 to one common marked fiber.

Inside R_* put

$$T_\alpha = c \rtimes_{\varphi_0} S^1_\alpha, \qquad T_\beta = e \times S^1_\beta.$$

Fix the basepoint $q = (p, *)$. Let $A = [\bar{\alpha}]^{-1}$ and $B = [\bar{\beta}]^{-1}$ be the based inverse base loops. Base the oriented meridians M, N along the initial fiber arcs y_1 to c_y and s_2 to s_e, respectively. In the product normal boundaries let λ_α be the y_1-whiskered push-off that lifts the inverse α direction and closes along the selected half of c, and let λ_β be the s_2-whiskered push-off of the inverse β direction along e. These are geometric boundary curves; their based words Ax and $(r^{-1}Mr)B$ are conclusions of Lemmas 4.3 and 4.4, not part of this definition. Define V_* by the two product-framed fillings

$$M\lambda_\alpha = 1, \qquad N\lambda_\beta = 1$$

in this fixed orientation sheet. Reversing an allowed peripheral orientation rewrites the same geometric fillings with exponents $(\epsilon_A, \epsilon_B) \in \{\pm 1\}^2$.

The definition singles out one manifold. The four sign labels in Theorem 2.1 are therefore convention sheets for this one object, not four nearby surgeries. The geometric work below has two logically distinct tasks: construct V_* from the simplicial model, and compare the marked smooth data of Definition 3.1 with the fixed member described by Wuebben. The first task is proved without the assumptions below. The second cannot be a theorem about a byte string alone, because it includes assertions about what another author's prose and figures denote.

3.1 Source-identification assumptions

Source-identification assumptions 3.2. The comparison with Wuebben's intended fixed member uses exactly the following four semantic assertions.

- (S1) **Marked-fiber fidelity.** Wuebben's imported marked fiber is the ordered five-chain (a, b, c, d, e) with the two marked complementary disks, chain-reversing orientation-preserving involution, and curve orientations used in Definition 3.1. In particular dashed arcs in the source figure depict hidden portions of the same curves, not additional crossings.
- (S2) **Marked-bundle fidelity.** The symbols φ_0 and $\psi_0 = T_a \circ T_b$ mean the supported right-twist representatives used here, with T_b applied first, relative to the full c and e collars; the target tori are the resulting c-over-α and e-over-β curve subbundles.
- (S3) **Peripheral and member fidelity.** The paper's symbols y_1, s_2, A, B, M, N denote the same oriented, commonly whiskered paths as in Definition 3.1; its fixed $(m, n) = (0, 0)$ member uses the two displayed filling slopes; and (ϵ_A, ϵ_B) changes only the allowed orientation sheet, not the geometric surgery.

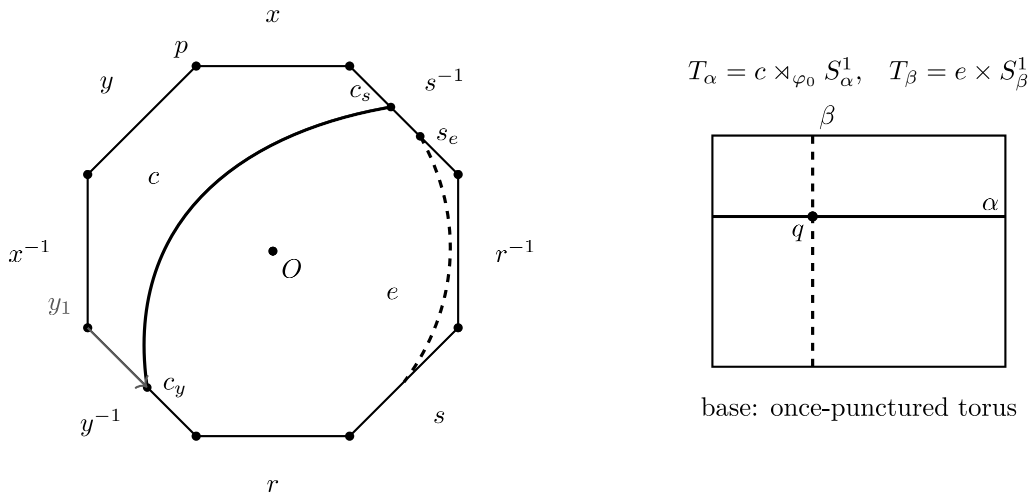
(S4) **Smooth-surgery fidelity.** Wuebben's smooth surgery tori are the images of those marked curve subbundles, and his geometric $+1$ Luttinger surgeries use their Lagrangian-framing classes with the orientations encoded by the relation sheet.

These are not placeholders for four missing computations. They are the four source-semantic statements at which the audit must say "this is what the target paper means." Table 2 records how far each has nevertheless been tested. The finite certificates, the construction of V_*, and the proof that $\pi_1(V_*) = 1$ do not use S1–S4.

Item	Finite or cited support	What is still assumed
S1	independent extraction from immutable source; exhaustive ribbon rotation check; periodic-disk theorem	intended identification of the source symbols and drawing convention
S2	relative collar checks, explicit graph clutching, based monodromy in two subdivisions, order/sign mutation controls	intended mapping-class and subbundle conventions
S3	two peripheral extractors; punctured beta sweep; complement-only certificates for both longitude words; author-script comparison	intended names, orientations, sign sheets, and selected family member
S4	constructive relative Moser normalization, two Weinstein-chart calculations, first-jet chart independence, Lagrangian-neighborhood theorem	intended smooth tori and surgery parametrizations

Table 2: The complete source-comparison boundary. The middle column is evidence for each reading; it is not silently promoted into proof of the semantic assertion in the last column.

The genus-2 fiber is realized as a regular neighborhood of the specified five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.



(a) the marked fiber F

(b) the two surgery tori

Figure 1: The marked data reconstructed by the audit. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent the basepoint p, while O is the center. Half-turn carries opposite edges to one another and induces $x \leftrightarrow r$, $y \leftrightarrow s$. The displayed arcs record the curve c, the five-chain curve e, and the initial y_1 whisker; the corresponding s_2 whisker is stored in the simplicial model. Panel (b) is a schematic of the two curve subbundles over the base loops. The geometric placements are checked as simplicial path and incidence assertions; the drawing is explanatory rather than an additional input to the computation.

The dictionary tested before Proposition 4.9 is displayed in Table 3; it is not reconstructed from the fact that the filled groups happen to collapse. Here "literal" means a stored edge path in the triangulation, with its basepoint and orientation retained.

Paper datum	Mathematical role	Certified realization
F, p, O	genus-two fiber with two fixed marks	86-vertex closed surface; both fixed vertices pass surface- and link-level tests
a, b, c, d, e	ordered filling five-chain	independently reconstructed ribbon; oriented c at V_2 represents $(rx)^{-1}$
φ_0	$x \leftrightarrow r$, $y \leftrightarrow s$	simplicial half-turn fixing p, O, checked on based paths
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}$, $y \mapsto yx$, $r \mapsto r$, $s \mapsto s$	ordered flip stack with T_b first; based monodromy replay
T_α	$c \rtimes_{\varphi_0} S^1_\alpha$	induced torus on the c-stack; local-flatness and subbundle checks
T_β	$e \times S^1_\beta$	induced product torus on the e-stack; local-flatness and product checks
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based edge paths from q, retaining their orientations
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers; all 592 normal fibers checked

Table 3: Marked-bundle dictionary for the fixed $n = 0$ member. Peripheral longitudes, which carry the sign-sensitive part of the identification, are displayed separately in Table 4.

Lemma 3.3 (equivariant marked fiber)**.** There is an orientation-preserving homeomorphism $h : |F_K| \to F$ carrying the ordered curves (a, b, c, d, e) and marked points (p, O) to those of Definition 3.1 and satisfying $h\varphi_K = \varphi_0 h$. It carries the named based loops x, y, r, s with their displayed orientations.

Proof. Finite evidence. The independent ribbon reconstruction has

$$i(a, b) = i(b, c) = i(c, d) = i(d, e) = 1, \qquad i(u, v) = 0 \quad \text{otherwise.}$$

Its regular neighborhood retracts to a graph with six vertices and ten edges, hence has Euler characteristic -4. The two disjoint symplectic pairs (a, b) and (d, e) force genus two, so the neighborhood has two boundary components. The complement in the certified genus-two surface has Euler characteristic two and exactly those two boundary circles; it is therefore two disks. An exhaustive rotation-system check begins with the 36 cyclic orders at the independent four-valent crossings. Alternation, orientation preservation, two boundary faces, and preservation of the two faces leave exactly four systems; reversing the permitted curve orientations normalizes all four to one equivariant oriented ribbon type. The primary and independent fibers realize that same labeled type, including the involution on every directed curve end and the two p/O faces.

Topological input. Map vertex disks and edge bands by the checked cyclic correspondence. These product maps agree on attaching intervals, so they give an equivariant homeomorphism of the ribbon neighborhoods. Each complementary disk is invariant: if the involution exchanged them it would have no fixed point there, contrary to the certified two-point fixed set. The classical periodic-disk theorem identifies each nontrivial orientation-preserving order-two disk action with a half-turn. Extending over the quotient disks and lifting gives the required equivariant marked map. The based path comparison then reads, literally,

$$\varphi_0 : (x, y, r, s) \longmapsto (r, s, x, y).$$

No fundamental-group filling or downstream theorem enters this argument. $\square$

Lemma 3.4 (relative monodromy and graph clutching)**.** The map h can be chosen so that it conjugates the two monodromy representatives relative to the full c and e collars. It consequently extends to an orientation-preserving fiberwise homeomorphism

$$H : |K| \longrightarrow (R_*)_{\mathrm{top}}$$

carrying the basepoint, both torus subbundles, and their product collars to the corresponding marked objects.

Proof. Finite evidence. The literal based actions are

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y), \qquad \psi_0 : (x, y, r, s) \mapsto (y^{-1}, yx, r, s).$$

The second action is obtained from complete whiskered paths, not free homotopy classes, and is replayed in two differently subdivided beta stacks. On the three-row c collar the alpha map is exactly the free shift by four columns. The beta representative is the ordered supported word $T_a \circ T_b$, with T_b first; all 1,536 trace cells avoid the full e collar, and the 3,072 restricted tetrahedra are exactly its product staircase.

Topological conversion. Both bundle models are the union of a common fiber block and two mapping cylinders over the rank-two spine of $\Sigma_{1,1}$; there is no base two-cell. For a monodromy f, write

$$M_f = (F \times [0, 1])/((z, 1) \sim (f(z), 0)).$$

If $hf_K = f_*h$, the formula $[z, t] \mapsto [h(z), t]$ respects the seam and has the analogous inverse. For an isotopic target representative, the standard interpolating formula $[z, t] \mapsto [k_t(z), t]$, with $f_*k_1 = k_0 f_K$, gives the same conclusion. The alpha conjugacy is exact on the c collar. For beta, both sides use the identical supported twist word; interpolating twist profiles is relative to the disjoint e collar. The two handle maps restrict to h on their common fiber and therefore glue, as do their inverses. This constructs H directly; bundle classification and Dehn–Nielsen–Baer are independent controls, not proof inputs. $\square$

Corollary 3.5 (the marked torus pair)**.** Under H, the induced subcomplexes $c \rtimes_{\varphi_K} S^1_\alpha$ and $e \times S^1_\beta$ map to T_α, T_β, together with their product collars and the p-section. All smooth and symplectic arguments may therefore be performed in the audit-defined smooth target R_*; no smoothing of $|K|$ is chosen.

The model results through Corollary 3.5 concern V_* and use no source-identification assumption. The paper-to-model comparison is exposed separately as an interpretation boundary rather than inferred from the later group collapse. Five structurally different constructions test subdivision, monodromy order, twist sign, labels, and orientations. A quarantined extractor checks the immutable Lidman–Piccirillo source and vector figure, and the frozen dictionary was compared diagnostically with Wuebben's scripts only after it was fixed; no proof input is derived from those scripts. Every compared entry agreed. Assumption 3.2(S1) records, rather than conceals, the surviving semantic convention that a dashed arc in the handle figure records hidden projection rather than an additional crossing. The complete comparison inventory and the S1–S4 audit are in the supplement and repository.

4 Peripheral data and the framing identification

Let $T = T_\alpha \sqcup T_\beta$ and let $C_* = R_* \setminus \operatorname{int} \nu(T)$. This section identifies the normal boundary of each component, its commonly based peripheral pair, and the two filling slopes. The statements are deliberately separated so that a failure of one cannot be hidden by triviality of the final groups.

Lemma 4.1 (local flatness and the normal frontier)**.** The two torus subcomplexes are disjoint, full, and locally flat. The barycentric level set

$$N_{1/2}(T) = \left\{ x : \sum_{v \in T} \lambda_v(x) \geq \tfrac{1}{2} \right\}$$

is an explicit normal D^2-block bundle, and its frontier $F_{1/2}(T)$ is the normal-circle boundary. The derived frontier used by the computation is PL homeomorphic to $F_{1/2}(T)$, carrying every extracted dual loop to a literal oriented normal-circle fiber.

Proof. For every simplex $\sigma \subset T$, the complete link pair is checked to be

$$(S^3, \text{unknotted } S^1), \quad (S^2, S^0), \quad (S^1, \varnothing)$$

in dimensions $0, 1, 2$, respectively. The enumeration covers all 296 vertices, 888 edges, and 592 triangles. Vertex-link 3-spheres and unknotted link circles carry independently replayed collapse and cyclic-knot witnesses. Thus the codimension-two local pairs are standard.

In a simplex with torus face A and opposite face B, barycentric coordinates give the unique block form

$$N_{1/2}(T) \cap \sigma \cong \Delta_A \times C(\Delta_B), \qquad F_{1/2}(T) \cap \sigma \cong \Delta_A \times \Delta_B.$$

The formulas agree on common faces. If $b(s)$ is a mixed-simplex vertex of the computed derived frontier, the map

$$b(s) \longmapsto \tfrac{1}{2} b(s \cap T) + \tfrac{1}{2} b(s \setminus T)$$

extends cellwise to the canonical subdivision map for $\Delta_A \times \Delta_B$ and hence gives the global PL homeomorphism. This is checked on 113,336 mixed vertices and 769,256 derived comparabilities. For every torus triangle the alternating dual loop has constant torus coordinate and is precisely the subdivided normal link circle. No regular-neighborhood uniqueness or smoothing theorem is used. □

Lemma 4.2 (complement presentation)**.** The induced subcomplex on the vertices outside T is a strong deformation retract of $|K| \setminus |T|$. Its maximal-tree edge generators and oriented triangle boundaries give a presentation of $\pi_1(C_*)$. The raw presentation has 99,863 generators and 321,702 relators; the sealed Tietze certificate reduces it to the 3-generator, 78-relator complement presentation in the source of Theorem 2.1, while transporting all 89 named peripheral and comparison words.

Proof. On each simplex outside the full subcomplex T, set the coordinates on T to zero and renormalize the remaining barycentric coordinates. The straight-line homotopy stays outside T, agrees on faces, and fixes the induced complement. A maximal tree on its 8,860 vertices has 8,859 edges; subtracting from the 108,722 complement edges gives the stated 99,863 generators.

Collapsing the tree and applying cellular van Kampen to every oriented two-simplex gives the raw presentation. Each of the 99,860 recorded Tietze eliminations is replayed against the unique current relator that implies it; separate Python and Ruby checker implementations agree on the final presentation and every tracked word. □

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction package (the corrected relations of its §8.3 and the push-off basing correction of its §8.5) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper's corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The closed section $\widehat{\Gamma}$ of the doubled bundle — the union of the two copies of the section through a fixed point of φ_0 — is extracted as a closed orientable genus-2 surface, and its self-intersection $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ is certified with an explicit normal push-off; this number enters the conditional consequences and the complete ledger in the supplement.

The relation sheet and the two fillings

To make Theorem 4.7 inspectable without opening the repository, fix the orientation sheet used by the triangulation and put $\delta = r^{-1}$. Before filling, the paper-coordinate presentation read from the literal paths has the surface relation, the following eight transport relations, and the drilled-fiber relation:

$$\begin{aligned}
&AxA^{-1} = r, \quad && AyA^{-1} = s, \quad && ArA^{-1} = x, \quad && AsA^{-1} = Ny,\\
&BxB^{-1} = y^{-1}, \quad && ByB^{-1} = M^{-1}yx, \quad && BrB^{-1} = r, \quad && BsB^{-1} = (r^{-1}M^{-1}r)s,\\
&[x,y][r,s] = 1, \quad && B(s^{-1}r^{-1}yx)B^{-1} && = r^{-1}s^{-1}x. &&
\end{aligned}$$

The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \qquad \lambda_\beta = (r^{-1}Mr)B,$$

and the four convention sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \qquad N\lambda_\beta^{\epsilon_B} = 1, \qquad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \tag{1}$$

Changing an allowed orientation inverts the corresponding meridian or displayed longitude and changes the sheet label; it does not define a new geometric surgery. The finite presentations of Theorem 2.1 are the frozen Tietze images of these literal path presentations, but that geometric assertion is part of Theorem 4.7, not of the finite theorem.

Direction	Paper word	Stored based loop	Independent evidence
λ_α	Ax	`lb_a_y1`, based by y_1 at c_y	literal path construction; Run 68's 2,506-record complement certificate
λ_β	$(r^{-1}Mr)B$	`lb_b_s2`, based by s_2 at s_e	punctured transport square and independent extraction; Run 69's 1,540-record complement certificate

Table 4: The two sign-sensitive longitude identifications. Neither certificate contains a filling relation. Reversing an allowed orientation inverts the corresponding displayed element and is recorded by (ϵ_A, ϵ_B).

Worked peripheral extraction

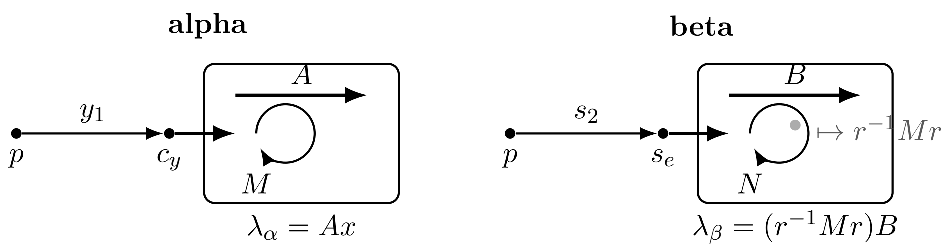


Figure 2: Common-whisker peripheral construction. On the alpha side the meridian and longitude use the same y_1 path to c_y. On the beta side they use the same s_2 path to s_e; the swept basing square meets T_α once, and puncturing that intersection contributes the conjugated meridian $r^{-1}Mr$. The diagram records basing and word order, not metric geometry.

Lemma 4.3 (the alpha peripheral pair)**.** Under the normal-frontier identification of Lemma 4.1, the stored loops `geom_M` and `lb_a_y1` are the commonly based meridian and product longitude (M, λ_α), and

$$\texttt{lb\_a\_y1} = Ax \quad \text{in } \pi_1(C_*).$$

The displayed equality is certified using only the 78 complement relators, with neither filling relation present.

Proof. Consider T_α. The curve c meets the oriented y edge once, at c_y. Starting at p, the stored path follows the initial segment y_1 to c_y, makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M. The longitude uses that identical whisker and normal jog, traverses the base direction A, and closes along the selected half of c. The stored words are `geom_M` and `lb_a_y1`; they are what M and λ_α denote throughout, and they are what Theorem 2.1 literally fills with.

That the two loops use the same whisker is not an inference but a fact one can read off the sealed transport input: there $\texttt{geom\_M} = y_1\, \mu_\alpha\, y_1^{-1}$ and $\texttt{lb\_a\_y1} = y_1\, \lambda_\alpha^\circ\, y_1^{-1}$ with the *identical* four-edge whisker $y_1 = e_{24046}e_{90328}e_{79763}e_{72645}$, seven- and six-edge circles $\mu_\alpha, \lambda_\alpha^\circ$, and fifteen and fourteen letters in all. So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the stored word `lb_a_y1` with the paper's symbol Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y, whiskered by y_1, jogged to the normal-block frontier, lifting the base generator A, and closed along the selected half of c. Algebraically, the elementary elimination pass over the sealed presentation leaves the 72-letter residual $\texttt{lb\_a\_y1}^{-1}Ax$, but a proof-producing completion reduces that exact word to the identity in 61 steps. Independent Python and Ruby verifiers replay the complete 2,506-record ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/alpha_residual/` (Run 68). An earlier run had reported an empty elementary residual over a presentation predating both committed exports. That evidence does not reproduce and is superseded by the sealed certificate just described. □

Lemma 4.4 (the beta peripheral pair)**.** Under the same normal-frontier identification, the stored loops `geom_N` and `lb_b_s2` are the commonly based meridian and product longitude (N, λ_β), and

$$\texttt{lb\_b\_s2} = (r^{-1}Mr)B \quad \text{in } \pi_1(C_*).$$

Again the equality is certified in the unfilled complement.

Proof. The beta-coordinate word $\texttt{lb\_b\_s2}^{-1}(r^{-1}Mr)B$ likewise leaves a 113-letter residual under elementary elimination. The same proof-producing complement system reduces that exact word to the identity in 82 steps; separate Python and Ruby verifier implementations replay its 1,540-record ancestry cone. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/beta_residual/` (Run 69). Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 2.1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e, the base loop B, and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the paper's word for this direction is $(r^{-1}Mr)B$ and not B. The stored words are `geom_N` and `lb_b_s2`, exported without substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ is certified in the complement exactly as in the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words "meridian" and "push-off." □

Lemma 4.5 (the drilled-fiber relation)**.** The based relation

$$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$$

holds in $\pi_1(C_*)$, not merely in the unpunctured mapping cylinder.

Proof. The transported loop κ_3, of edge length 31, misses c but meets e once; sweeping it along the core of the beta curve would therefore meet T_β and would not prove the stated complement relation. The paper uses the specified parallel push-off. Along that push-off, the complete mapping-cylinder stack — 17,664 tetrahedra on 2,966 vertices — has no vertex on either surgery torus. A 2×35 simplicial homotopy grid at p joins the push-off basepoint loop to the core one and identifies it with the presentation's whiskered generator; it too is disjoint from both tori. The based monodromy target then reduces to $r^{-1}s^{-1}x$.

The finite checker exhausts the stack, both torus vertex sets, every grid cell, and the target path. Thus the entire transport homotopy lies in C_*. This is the construction-specific fact that was absent from the earlier mapping-cylinder-only calculation; no group simplification or low-index fingerprint is used to supply it. □

Theorem 4.6 (product framing equals Lagrangian framing)**.** For the normalized Thurston symplectic form on R_*, the product-framed push-offs in Lemmas 4.3 and 4.4 represent the canonical Lagrangian-framing classes of T_α and T_β. In particular neither acquires a meridian component when transported into a Weinstein neighborhood.

Proof. The proof takes place in the already smooth target R_*. By Corollary 3.5, the marked homeomorphism H transports the product collars and their push-offs from the PL model into that target. No smooth structure is placed on the source complex. A framing push-off below means the isotopy class, in $\nu(T_i) \setminus T_i$, of a sufficiently small copy of a curve on T_i.

Relative normal form. On an annular fiber collar write

$$\Omega = f(\theta, t)\, dt \wedge d\theta, \qquad f > 0,$$

with core $t = 0$. Put

$$g(\theta, t) = \int_0^t (f(\theta, u) - 1)\, du, \qquad H_s(\theta, t) = t + sg(\theta, t).$$

Then $\partial_t H_s = (1 - s) + sf > 0$. On one collar common to the compact core, H_s therefore has a smooth inverse in the radial coordinate. If $H_s(\theta, T_s(\theta, t_0)) = t_0$, then $T_0 = t_0$, $T_s(\theta, 0) = 0$, and

$$\partial_s T_s = -\frac{g(\theta, T_s)}{(1 - s) + sf(\theta, T_s)}.$$

Thus $(\theta, t_0) \mapsto (\theta, T_s)$ is the relative Moser isotopy, obtained by monotone inversion rather than an abstract flow theorem. At $s = 1$, differentiation in t_0 gives $f(\theta, T_1)\partial_{t_0} T_1 = 1$, hence $T_1^*\Omega = dt_0 \wedge d\theta$. The exact identities, fixed-core condition, and a uniform collar bound are replayed in `moser_cumulative_flow.py`.

For the alpha collar the deck involution is $\tau(\theta, t) = (\theta + \pi, t)$ and the quotient coordinate is $\bar\theta = 2\theta$. The preceding radial map is defined directly upstairs by replacing $\bar\theta$ with 2θ in its coefficient. Periodicity makes it τ-equivariant, so no covering-lift choice is involved. Moreover

$$q^*(dt \wedge d\bar\theta) = 2\, dt \wedge d\theta = d(2t) \wedge d\theta,$$

which is the required factor-two normalization. These assertions are replayed in `equivariant_moser_lift.py`.

The two charts. Near T_β, ψ_0 is the identity near e, and the preceding normalization gives

$$\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2;$$

the cotangent momenta are (t, Ku). The in-fiber and base-normal push-offs are therefore constant-momentum copies. Near T_α, the mapping-torus seam is

$$(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u).$$

On its quotient put

$$\Theta_1 = \theta_1 - \tfrac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \tfrac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K\, du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The Liouville primitive $t\,d\theta_1 + Ku\,d\theta_2$ forces these momenta by pairing with the Θ-frame; `weinstein_chart_check.py` replays that derivation, the seam, exterior algebra, and a second opposite-shear chart.

Independence of the Weinstein chart. It remains to show that the constant-momentum conclusion is a framing class rather than a feature of the displayed coordinates. Given two Lagrangian-neighborhood charts, first compose with a cotangent lift so that their restrictions to the zero section agree. Their transition is then a symplectomorphism germ $F : (T^*T, T) \to (T^*T, T)$ fixing the zero section pointwise. With fiber dilation $\delta_r(q,p) = (q, rp)$, define

$$F_r = \delta_r^{-1} F \delta_r \quad (0 < r \leq 1).$$

Although $\delta_r^* \omega = r\omega$, the two factors cancel, so every F_r is symplectic. Along the zero section the symplectic condition gives

$$dF = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix}, \qquad A = A^{\mathrm{T}}.$$

Taylor expansion shows that F_r extends at $r = 0$ by the identity. Hence F_r is an isotopy of germs, relative to T, from the identity to the chart transition. Applied to a sufficiently small nonzero constant-covector copy of either named curve, this compact isotopy stays in $\nu(T) \setminus T$ and proves that the two push-offs have the same boundary class. The first-jet symplectic calculation is replayed in `weinstein_chart_independence.py`. The potentially dangerous momentum shift by a closed one-form could alter the pushed-off boundary class, but it moves the zero section unless the form vanishes and is therefore excluded here. The standard Lagrangian-neighborhood theorem, in the form used by [3, §2.1 and Proposition 2.2], supplies the existence of the charts; the argument above supplies the precise chart-independence consequence needed here. □

Theorem 4.7 (audit-model identification)**.** For each coherent convention sheet $(\epsilon_A, \epsilon_B) \in \{\pm 1\}^2$, there is a natural surjection

$$P_{\epsilon_A, \epsilon_B} \twoheadrightarrow \pi_1(V_*).$$

In fact the map is an isomorphism. The four sheets express peripheral orientation conventions for the one audit-defined manifold V_*, not four geometric targets. No source-identification assumption S1–S4 enters this statement.

Proof. Lemmas 3.3 and 3.4 construct the marked bundle homeomorphism and carry the two curve subbundles and their product collars to the specified tori. Lemma 4.1 identifies their normal boundaries and meridians. Lemmas 4.3 and 4.4 identify the two commonly based product longitudes, including the conjugated meridian contributed by the punctured beta sweep. Lemma 4.5 places the final transport relation in the drilled complement. Theorem 4.6 identifies the product push-offs with the Lagrangian-framing push-offs.

Consequently attaching the two Luttinger solid tori adds exactly the normal closure

$$\langle\langle M\lambda_\alpha^{\epsilon_A}, N\lambda_\beta^{\epsilon_B}\rangle\rangle.$$

Cellular van Kampen therefore gives the asserted surjection from the filled presentation. Lemma 4.2 gives an actual presentation of the complement, not merely a list of valid relations; applying van Kampen with the two filling relations proves that the surjection is an isomorphism. Changing a common whisker simultaneously conjugates both elements of a peripheral pair, while changing an allowed orientation replaces the corresponding displayed exponent. Neither operation changes the filled manifold. This completes the identification with Definition 3.1 without comparing that definition to the intended denotation of Wuebben's symbols. □

Corollary 4.8. The audit-defined manifold V_* is simply connected.

Proof. Use any convention sheet in Theorem 4.7. Its source is trivial by Theorem 2.1; a quotient of the trivial group is trivial. Notice that this conclusion uses only the surjective part of Theorem 4.7 and no source-identification assumption. □

Proposition 4.9 (conditional comparison with Wuebben)**.** Assume S1–S4 of Assumption 3.2. Then the marked smooth surgery data defining V_* agree with the data of Wuebben's fixed $(m, n) = (0, 0)$ member. In particular the resulting filled manifolds are orientation- preservingly diffeomorphic, with their peripheral and framing data identified.

Proof. By S1, Lemma 3.3 identifies the ordered marked fiber and involution. By S2, Lemma 3.4 and Corollary 3.5 identify the relative monodromies, bundle, tori, and product collars. By S3, Lemmas 4.1, 4.3, 4.4, and 4.5 identify the commonly based meridians, longitudes, transport relation, sign sheets, and selected family member. By S4, Theorem 4.6 identifies the product push-offs with the smooth Lagrangian-framing classes used in the two $+1$ surgeries. These marked identifications carry both attaching maps and therefore the filled manifold of Definition 3.1 to the fixed member named in the target paper. The source-extraction and mutation checks in Table 2 are evidence for S1–S4, but are not used here as proofs of authorial intent. □

Corollary 4.10. Assuming S1–S4, Wuebben's intended fixed member is simply connected.

Proof. Combine Corollary 4.8 with Proposition 4.9. □

5 Conditional manifold consequences

The following is a relative theorem. Its difficult inputs are published classification, symplectic, and Floer-theoretic results; the machine ledger checks dependency bookkeeping and evidence hashes, not those theorems.

Theorem 5.1 (conditional consequences)**.** Assume S1–S4 and the twenty-five external results enumerated in the companion supplement and their stated applications. Then Corollary 4.10 implies Wuebben's three conclusions:

(A) $Z = V_* \cup_\sigma V_*$ is homeomorphic but not diffeomorphic to $S^2 \times S^2$;

(B) $W = V_*/\sigma$ is homeomorphic to Kawauchi's manifold B, while (B, W) is a homeomorphic non-diffeomorphic pair distinguished by smooth sliceness of the figure-eight knot;

(C) $Z'' = V_* \cup_{f\circ\sigma} V_*$ is homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Dependency audit. For (A), van Kampen and Corollary 4.10 make Z simply connected. The Euler-characteristic, signature, spin, and square-zero section calculations give the even hyperbolic intersection form, so Freedman's classification gives the homeomorphism type. The Thurston form and Theorem 4.6 make the surgeries symplectic. Ho–Li invariance of symplectic Kodaira dimension, together with the asphericity and minimality inputs recorded in the supplement, excludes the standard smooth $S^2 \times S^2$.

For (B), the regular double cover has deck group $\mathbb{Z}/2$, and the recorded homology, spin, intersection-form, and Kirby–Siebenmann data meet the Hambleton–Kreck hypotheses, identifying W topologically with B. The figure-eight knot bounds the constructed disk in W. If it bounded smoothly in B, the trace embedding and Klug–Levine Arf calculation would give the prohibited congruence $\Delta_{4_1}(-1) = -5 \not\equiv \pm 1 \pmod 8$.

For (C), the regluing calculation and Corollary 4.10 give a closed simply connected manifold with the odd unimodular rank-two form, hence the homeomorphism type $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. The Lidman–Piccirillo regluing and Floer-theoretic inputs give a nonzero mixed invariant, while the connected sum has vanishing mixed invariant; the orientation adjustment is made before that comparison. This proves non-diffeomorphism relative to those cited inputs.

The frozen dependency certificate contains 25 external theorems, 3 project certificates, 15 recomputed facts, and 16 deductions. Two separate implementations verify its acyclicity, evidence digests, and the dependence of all three conclusions on Corollary 4.10. The full statements and hypothesis ledger are retained in the supplement so that this bookkeeping does not receive the same visual weight as the audit-model identification and the separate S1–S4 source comparison. □

6 Limitations and the unresolved contrary computation

Wuebben's public repository has stated continuously from its first visible commit that Lidman and Piccirillo report a fundamental-group computation disagreeing with his; the current wording reports $\pi_1(V) \neq 1$. Their published paper [2] makes no such assertion: it proves $H_1(V) = 0$ and normal generation by $\pi_1(F)$, and explicitly does not fix a unique surgery parametrization. The contrary calculation is therefore a correspondence report relayed by Wuebben. No presentation, relation sheet, generator dictionary, or nontriviality witness for it is public in the materials examined here.

It is consequently unknown whether the two computations concern the same member of the parametrized family, and equally unknown that they concern different members. The finite Theorem 2.1 is unaffected: its objects are explicit strings. A contrary computation of the same fixed member would instead bear on Proposition 4.9 and Assumption 3.2, where this paper locates the construction-specific risk. Relation-by-relation reconciliation requires the missing artifact. The repository contains a dated inventory of the public search and is ready to ingest such an artifact if supplied.

The project has two further limitations. First, both theorem-critical filled-group checkers were produced within the same AI-assisted development process. Separate languages and implementations, negative controls, and the published mathematical specification reduce ordinary software risk but do not constitute an independent human line-by-line audit. Second, the marked fiber, peripheral, and framing arguments have been decomposed above so that a PL or symplectic topologist can audit them locally, but no such external audit is claimed. The transfer from V_* to Wuebben's member is expressly conditional on S1–S4; Theorem 5.1 is additionally conditional on its cited external results and on the semantic correctness of their applications.

The 100-case framing scan is not an answer to either limitation. All sampled shifted presentations were reported trivial, so the scan does not discriminate the correct framing and has no logical role in Theorem 4.7 or Proposition 4.9. It is retained only as computational history in the supplement.

7 Conclusion

The revision isolates two theorem-level achievements. First, four hash-identified presentations are trivial under a complete published certificate specification, with the raw transport and all retained derivations replayable from frozen inputs. Second, those presentations identify the fundamental group of the explicitly defined V_*, which is therefore simply connected. The marked fiber, relative clutching, torus collars, normal frontier, complement presentation, two peripheral pairs, drilled-fiber relation, and Lagrangian framing are separate results whose finite and conventional inputs are visible.

The remaining construction-specific uncertainty is S1–S4: whether the source symbols and diagrams denote exactly the marked data defining V_*. The paper-to-V_* transfer and the unavailable contrary computation meet at that boundary. The artifacts are released to make those assumptions finite and inspectable, not to promote source interpretation into a certificate.

8 Data and reproducibility

The target is Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`.

The immutable Lidman–Piccirillo v1 source digest and all interpretation inputs are recorded in `TARGET.md`. The theorem artifacts, frozen inputs, certificate checkers, manifests, run transcripts, and reconstruction programs are in the public repository

`https://github.com/VentiMath/exotic-s2xs2-verification`.

The submission release has three intentionally different replay levels. The certificate-only path reads frozen files and requires only standard Python and Ruby runtimes. The geometric-audit path recomputes the finite marked-fiber, torus, frontier, peripheral, and framing checks. The full reconstruction path additionally uses the locked GAP 4.11.1 and

KBMAG 1.5.9 environment and the proof-producing KBMAG patch. Commands, expected record counts, negative controls, and hashes are listed in the companion supplement. The deterministic source and archive digests are recorded in `ARXIV_SUBMISSION.md`.

This revision is prepared for repository release `v2.0.0`; its release gate requires the public tag, paper and supplement PDFs, deterministic source archives, manifest, and clean-checkout replay to agree before circulation. A DOI-bearing archival deposit is still required before journal submission; until that deposit exists, the versioned release and hashes are the precise reproducibility target.

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